Efficient CNN Architectures for Medical Image Classification: Reproducing and Evaluating Eff-PCNet

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UCSD PHYS 139

Abstract

Deep learning, Transformer and MLP based methods have achieved strong performance on medical image classification tasks. But, these methods have several challenges when implementing them in medicine science. Limited computational resources and datasets pose significant challenges. The Eff-PCNet architecture was proposed as an efficient CNN that combines multi scale depthwise convolutions and structural re-parameterization to decrease computational costs while maintaining performance on medical image datasets. In this project, we implement a transfer learning baseline using EfficientNetV2-S and a lightweight EffPCNet-Small model inspired by Eff-PCNet. We evaluate both models on three datasets including HAM10000 (7 class dermoscopy), SkinCancer MNIST (benign vs malignant), and ChestXray Pneumonia dataset (normal vs pneumonia). Our EffPCNet-Small model is not an exact replica of the paper's architecture, it follows the same high level design by incorporating M2C and Rep-C style modules, but uses reduced depth, simplified channel splits, and omits inference time structural re-parameterization due to resource constraints and memory limits. Despite these constraints, EffPCNet-small significantly improves accuracy, F1, and AUC compared to the EfficientNetV2 baseline on SkinCancer and HAM10000 with roughly a fifth of the parameters. On ChestXray, EfficientNetV2 achieves higher accuracy, but EffPCNet-Small attains a higher AUC. Overall, our results reproduce the qualitative trends of the Eff-PCNet paper as the EffPCNet style architecture provides comparable performance at much lower computational costs. Our code can be found here:
https://github.com/davebobculver/Phys_139_project/tree/main
OR
https://colab.research.google.com/drive/1tSDXoeicGtcOS97Lz0Jh_QbIYUjpacNh?usp=sharing

1. Introduction

Medical image classification is an important diagnostic tool, with applications ranging from skin lesion triage to pneumonia detection. However, many high performing models are computationally expensive. This limits practicality and real time deployment, a challenge analogous to applying ML models in high throughput physics experiments where latency and resources are strict (think CERN LHC).

The Eff-PCNet architecture was proposed as an efficient CNN that aims to address the above trade off by combining multi-scale convolutions and structural re-parameterization to reduce redundancy while maintaining strong performance on several medical imaging benchmarks. The authors introduce two key modules. First, the M2C (Multi-branch Multi-scale Convolution) module, which approximates very large receptive fields using multiple depthwise convolution branches. Second, the REP-C module, which uses structural re-parameterization to combine multiple convolutional branches into an equivalent convolution at inference.

Our project's main goals are: First, implement a strong baseline using EfficientNetV2-S with ImageNet pretraining. Next, implement a lightweight EffPCNet-Small model inspired by Eff-PCNet's M2C, and Rep-C ideas to fit within our current RAM constraints. Last, evaluate these models on three medical datasets and compare accuracy, F1, AUC, as well as parameter counts, and qualitative behavior to the trends reported in the Eff-PCNet paper.

2. Background and Related Work

2.1 CNNs, Transformers, and medical imaging

CNNs are the standard approach for image classification, including medical imaging tasks. Architectures like ResNet and EfficientNet family models leverage depthwise convolutions and other methods to improve accuracy, but at the same time increasing computational cost.

2.2 Eff-PCNet

Eff-PCNet aims to build an efficient CNN by introducing new modules that replace large kernels and heavy multi branch blocks with more efficient structures. M2C module: Splits feature channels into four branches: three depthwise convolution branches (3x3, 1xK, Kx1) and an identity branch. Channels are split on a ratio of a = 0.375, where the larger portion goes to the identity. With the model architecture this corresponds to 24 and 40 for convolution and identity respectively. The 1xK and Kx1 branches approximate large KxK kernels with lower computational cost. The outputs are concatenated and compressed with a 1x1 convolution. This achieves multi scale receptive fields without large parameter overhead.

MBC (Multi Branch Convolution) blocks: Use multiple convolutional paths and squeeze-and-excitation to reduce redundancy to emphasize informative channels.
Rep-C module: During training, multiple convolutional branches are used. During inference, weights are combined into a single convolution reducing latency and preserving representation.

3. Methods

3.1 Datasets

We use three publicly available, medical image datasets, downloaded via kagglehub where we then unpacked and reorganized. HAM10000: 10,015 dermatoscopic images labeled into 7 classes of skin lesions. SkinCancer MNIST: 3297 skin lesion images labeled as either benign or malignant. Chest X-ray Pneumonia: 5856 X rays labeled as normal or pneumonia. We construct a non overlapping 80/20 train/validation split for each dataset. This ensures a fixed split and prevents data leakage.

3.2 Preprocessing

All images are resized to 224x224 and loaded with RGB channels. Then, apply data augmentation on the training set: Random horizontal flip, Random rotation (+/-0.05 radians), Random zoom (10%), Random contrast adjustment. Then, we apply normalization: Rescale pixel intensity by 1/255 to [0,1]. These are implemented as a small Keras Sequential preprocessing model and applied via Dataset.map.

3.3 Baseline model: EfficientNetV2-S

Our baseline is a transfer learning classifier based on EfficientNetV2-S. We load in ImageNet weights and keep the baseline model frozen so only the new head is trained. On top of the pooled features, we add global average pooling, a dropout layer of 0.3, and a dense softmax layer. This baseline model, labeled EffNetV2-baseline, represents a strong CNN baseline similar to the one in the Eff-PCNet paper.

3.4 EffPCNet-Small (ours)

We implement a lightweight EffPCNet-Small architecture that captures the core ideas of Eff-PCNet but constrained for our computational resources and memory limitations. The backbone layout is a custom CNN with a stem followed by stages. The stem is a several Convolution-Batch Normalization-Swish Activation blocks with increasing channels (24, 32, 48, 64) and downsampling. Stage 4 is a M2C style block that splits channels into 4 branches (3x3 depthwise, 1xK depthwise, Kx1 depthwise, and identity), concatenates, and applies a 1x1 convolution + batch normalization + ReLU. Stages 5, 7, 8 are simplified Rep-C style multi branch blocks with 3x3, 1x1, and depthwise 3x3 paths plus a shortcut and SE. Next is the head which consists of global average pooling, dropout (0.3), and dense softmax classifier. Some important limitations to be noted is firstly that we use fewer repeated blocks and sometimes smaller channel widths than the full Eff-PCNet architecture to keep memory usage low. Secondly, our Rep-C block is multi branch only during training and inference as we do not implement a separate inference time structural re-parameterization step that combines weights into a single kernel as done so in the paper. Overall, our custom model has roughly 4.39M parameters, far less than EfficientNetV2-S's 20.3M while still implementing paper ideas.

3.5 Training and metrics

All models are trained using Adam optimizer with learning rate 1×10^{-4}, categorical cross entropy (one hot labels) loss, batch size 32, 20 epochs, no early stopping. We evaluate using Accuracy, F1 score, AUC, and confusion matrices.

4. Results

4.1 EfficientNetV2-baseline

Table 1: EfficientNetV2-baseline validation results

Dataset	Acc	F1	AUC
SkinCancer	0.6449	0.6312	0.7445
HAM10000	0.6755	0.1165	0.5583
ChestXray	0.8275	0.7212	0.8700

On the binary ChestXray dataset, the baseline performs quite well, with ~83% accuracy and solid F1 and AUC. On SkinCancer, performance is moderate. On HAM10000, the F1 and AUC are poor despite ~67% accuracy. Looking at confusion matrices, we see the model tends to overpredict one particular class, suggesting class imbalance issues.

4.2 EffPCNet-Small (ours)

Table 2: EffPCNet-Small validation results

Dataset	Acc	F1	AUC
SkinCancer	0.8498	0.8494	0.9266
HAM10000	0.7668	0.4589	0.9226
ChestXray	0.6362	0.6269	0.9721

EffPCNet-Small significantly improves performance on SkinCancer as accuracy increases by ~20%, F1 and AUC rise. On HAM10000, accuracy improves moderately while F1 and AUC dramatically improve, indicating the model better separates classes. On ChestXray, the model has lower accuracy than baseline, but AUC is higher, suggesting that our model may have a higher false negative rate.

4.3 Efficiency comparison

EfficientNetV2-baseline has ~20.3M parameters while EffPCNet-Small has ~4.39M parameters. With fewer parameters and shallower depth, our custom model trains faster and uses less memory. Combined with improved performance on SkinCancer and HAM10000, this supports the main claim of Eff-PCNet that multi-scale and multi branch CNN modules offer a better performance to computational cost trade off.

5. Discussion

Our results reproduce the qualitative trends seen in the Eff-PCNet paper, but at a non exact performance due to architectural and training differences. On SkinCancer and HAM10000, EffPCNet-Small consistently outperforms the EfficientNetV2 baseline in F1, AUC, and accuracy. These gains are especially significant on HAM10000, which is highly imbalanced and contains 7 fine grained lesion classes. The M2C and multi branch blocks appear to help capture subtle patterns across scales, which is valuable for differentiating visually similar lesions. On ChestXray, the EfficientNetV2 baseline achieves the best accuracy, while our EffPCNet-Small achieves a higher AUC. This suggests EffPCNet-Small has good separability but may require threshold re calibration due to higher false positive rates to match the accuracy of the baseline. We do not match the exact metrics reported in the paper due to various reasons. First, our custom model has fewer repeated blocks and some smaller channels than the paper's architecture. Second, we train for 20 epochs instead of 300 as the paper did which may under train both models. Third, we use a random 80/20 split instead of the paper's specific ID split. Last, our Rep-C block is multi branch but does not combine branches into a single kernel at inference, thus missing the benefits of structural re-parameterization. However, our model still achieves better performance than the baseline on two datasets with roughly ⅕ the parameters.

6. Conclusion and Future Work

In this project, we implemented and evaluated two convolutional architectures on three medical image datasets, an EfficientNetV2-S transfer learning baseline and a custom lightweight EffPCNet-Small inspired by the paper's Eff-PCNet. Our custom model incorporates key concepts such as M2C style multi scale convolutions and multi branch residual blocks and achieves substantial improvements over the baseline on SkinCancer and HAM10000 with fewer parameters. On ChestXray, our model traded accuracy for higher AUC. Future work could focus on implementing the full architecture, training for more epochs with scheduling, implementing true structural re-parameterization, and applying class balanced sampling for HAM10000. In conclusion, our results support the paper's claims, delivering performance gains at lower computational costs.

Works Cited

Yue, W., Liu, S., & Li, Y. (2023). Eff-PCNet: An Efficient Pure CNN Network for Medical Image Classification. *Applied Sciences, 13*(16), 9226. https://doi.org/10.3390/app13169226